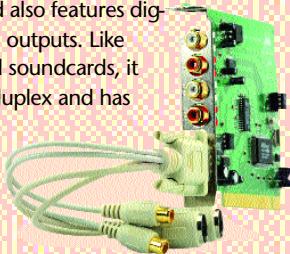


Professional Hardware

M-Audio Audiophile 2496 Soundcard:

While the SoundBlaster series from Creative is heralded as a high-end soundcard series, its features and functions appeal mainly to the gamer. M-Audio is a professional audio company that has built a reputation among musicians looking for professional-grade digital audio. The Audiophile 2496 features full 24-bit and 96 KHz audio support, aimed at multi-track recording. The card is fitted with two analog inputs and outputs (one stereo pair each) and also features digital inputs and outputs. Like most pro-level soundcards, it supports full duplex and has multi-client support, which means that more than one program can access the soundcard simultaneously.



Shure SM58 Microphone:

The Shure SM58 dynamic microphone is perfect for all kinds of vocal applications, making it a favourite among vocalists and sound professionals. This mic boasts of excellent sound feedback and extended frequency response from 50 to 16,000 Hz. Its rugged construction with a hardened steel mesh makes it an investment that will last you for a long time.



ATI 8MX2 Mic Preamp/Mixer: If you plan on recording from more than one sound source at a time, you need a mixer which ensures that everything gets to the soundcard at a proper line level, while allowing you to leave multiple sources plugged in simultaneously. The 8MX2 is an impressive digital multi-track system that offers eight mic preamps and a stereo mix bus with good monitoring capabilities.

Studiophile Monitors: 'Monitors' is a term used for the speakers that let musicians hear what they play while they play it. You can have the best gear in the world and still have no idea how it's really going to sound unless you have monitors that let you listen to your tracks with faithful reproduction. With M-Audio's Studiophile line of studio monitors, even the gentlest nuances reach your ears. The most popular from this line are the SP-8B speakers with 8-inch woofer or the smaller, bookshelf-sized SP-5B with 5-inch woofer. Add the optional SP-8S subwoofer to either model and you have all the bass you would need during your recordings.



Here are some possible scenarios, depending on the kind of gear accessible and whether you want to record your music all at once or each instrument individually. If you have a sequencing software, you can individually record parts of the instrumental and vocal sections and arrange and edit them later. This is especially useful if you are just getting into audio recording and are not a 'one-take' musician.

Scenario 1: Spontaneous sounds This is the most rudimentary method of recording audio. Here, your band plays as if delivering a live performance and all the vocal mics, drum mics, guitars, bass, and other instruments are fed into a single soundboard from where you can control the

volume and equalisations of each instrument. All these audio inputs are finally downmixed into a conventional stereo output which the soundcard records as a stereo file. You can control the levels of the musical instruments and vocals only during the recording and not afterwards.

Once you have a line level signal from the soundboard going into your soundcard, fire up the audio recording software, say SoundForge. At this point, you are basically using the computer as a simple digital tape recorder. You'll end up with a single stereo mix of the entire band, which is more or less a finished product. The only editing you can do is to equalise it, add basic effects such as Echo and Reverb or eliminate noise in the audio file.

You can't re-record just the rhythm guitar or replace the bass with a piano. This type of setup is suited to recording just the vocal track of a song or in cases where you essentially want to only change the audio characteristics of the recording such as volume, equalisation and noise filtering. The advantage is that the cost of equipment is relatively lower than setting up a multi-channel recording station.

Accessories you need: A full duplex soundcard; a rudimentary soundboard or mixer (M-Audio Audiobuddy; Price: \$160/approx Rs 8,000)

Scenario 2: Total control over individual tracks Here you record each instrument individually and do all the

Jargon Buster

Full duplex: The ability to simultaneously record and playback through the same audio hardware. Older soundcards (Half duplex) could either record or playback and were unsuitable for professional applications.

MIDI: An acronym for Musical Instrument Digital Interface, this format uses codes to define the type of sound a musical note plays. MIDI has no audio properties; it's just a set of instructions that are decoded by the audio hardware (soundcard, synthesizer, etc). Since actual sound data is not captured by this format, MIDI files are far

smaller than WAV files.

Patch: A MIDI term that describes the type of musical instrument being played by the MIDI synthesizer such as a tuba or a bass horn or a snare drum.

Polyphony: A term for the number of simultaneous voices or instruments that a synthesizer can play.

Soundboard: Also known as a mixer, it combines a number of input sources, allowing you to add effects and equalisation before sending the total mix to one or more outputs.

WAV: This format stores digital audio by a

process called sampling. Here, the audio signal is encoded by grabbing segments of it and encoding it in a time-based manner. For example, for CD quality audio, WAV files are captured at a frequency of 44.1 KHz in 16-bit samples.

Wavetable: A table of digitised samples of actual recorded sound that is stored in read-only memory (ROM) on a soundcard chip or in the main memory of a computer. This method is implemented in MIDI audio. Wavetable is much higher in fidelity and authenticity compared to FM synthesis.